K.R. MANGALAM UNIVERSITY

THE COMPLETE WORLD OF EDUCATION

AI IN CYBER SECURITY

(ENSP305)

B. TECH CSE

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1 Introduction to AI and Cyber Security

* Cyber Threat Landscape Analysis

→ Cyber threat landscape Analysis means studing the difference types of cybersecurity threats, attackers, vulnerabilities and that can effect computers, applications organizations and users

* In Simple Words:

→ It is the process of understanding "what cyber threats exists, who create them, how they attack and how we can protect

* What is cyberthreat?

- Steal information
- Damage Systems
- Disrupt Services
- Extort money

* Common cyber threats

	Threat	Meaning	Example
→	Phising	Fake message	Fake otp
→	Malware	Malicious software	virus, spyware
→	Pass. attack	Attempt to obtain pass..	Brute force attack
→	DDOS. Attack	overload a server	website become unav.

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* Who are the attackers?

→ 1. Cyber criminals :- usually motivated by money

2. Hacktivist:- Attack for political cause

3. Insiders:- Employee or other unauthorize users

4. Nation-Stage group:- Conduct cyber operations for govt.

* What does Threat Landscape Analysis Study?

→ A security analysist generally examines

Assets → vulnerabilities → Threat → Attack Methods → Risk → Protection.

* Main objectives

→ identify potential threats
→ Find system vulnareblities
→ understand attackers technique
→ Estimate potential impact
→ Pratrize security measures
→ Improtve incident response

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* How cyber threats Landscape Analysis Work.

Identify Assets
↓
Identify vulnereblities
↓
Identify potential attacker
↓
Analyze risk
↓
Analyze Potential ~~risk~~ Impack
↓
Apply Security Control
↓
Monitor continuesly

* Threat vs vulnerablity vs Risk

Term	Meaning	Example
Threat	Potential Source of harm	cybercriminal
vulnerablity	weakness that exploited	Weak password
Risk	Potential loss or damage resulting from thread exploiting vulnerablity.	Account comprom